

Nonlinear manifold approximation using Compositional Polynomial Networks



Antoine Bensalah¹ Anthony Nouy² Joel Soffo^{1,2}
¹Airbus, France
²Centrale Nantes, Nantes Université, LMJL UMR CNRS 6629

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Context

We are interested in the approximation of a subset $M = \{u(\mu) : \mu \in \mathcal{P}\}$ of a Hilbert space X . M can be for example the set of solutions of a parameter-dependent equation, with $\mathcal{P}$ being the parameter space. The goal is to approximate M by a n -dimensional manifold $M_n = \{D(a) : a \in \mathbb{R}^n\}$, with $a = E(u)$, where $E : M \rightarrow \mathbb{R}^n$ is the encoder, and $D : \mathbb{R}^n \rightarrow X$ is the decoder.

Keywords: Model order reduction, compositional polynomial networks, manifold approximation, sparse polynomials, low-rank polynomials;

State-of-the-art

Let $\phi_1, \dots, \phi_N$ be a basis of a linear space X_N .

Linear approximation

$$D(a) = \sum_{i=1}^N a_i \phi_i, \quad a_i = (u | \phi_i)$$

- Examples : PCA, POD, ...
- We may need **N large** to have $u \approx D(a)$.
 (Example : transport-dominated problems).

Nonlinear approximation idea

$$D(a) = \sum_{i=1}^n a_i \phi_i + \sum_{i=n+1}^N g_i(a) \phi_i.$$

The functions $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are nonlinear maps.

- Quadratic polynomials. [2, 3]
- Sums of univariate high-order polynomials. [4]
- Neural networks. [5]

An illustrative example

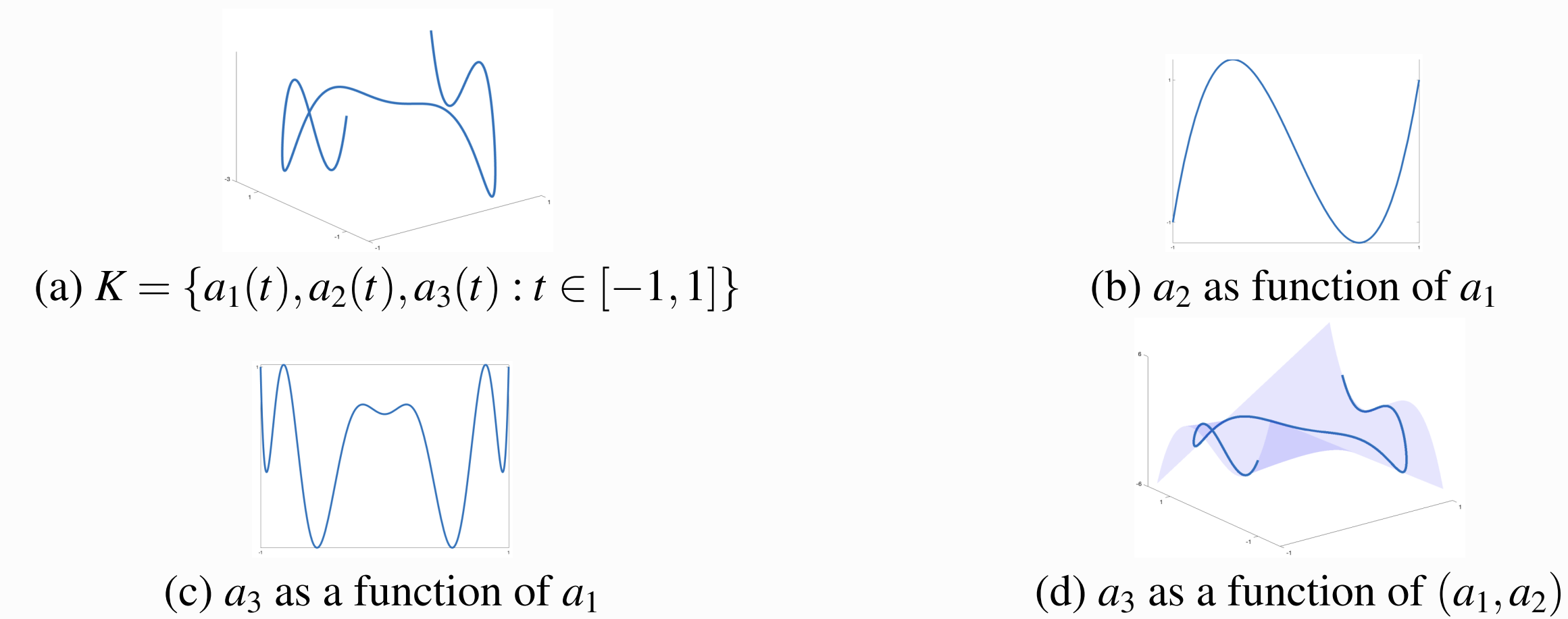


Figure 2: For $u = (a_1, a_2, a_3)$, instead of $u \approx (a_1, g_2(a_1), g_3(a_1))$, we do $u \approx (a_1, g_2(a_1), f_3(a_1, g_2(a_1)))$. f_3 is smoother than g_3 .

Numerical results

The functions f_i are sparse polynomials in these experiments.

KdV equation

$$\frac{\partial u}{\partial t} + 4u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0; M = \{u(\cdot, t) : t \in [0, 1]\}$$

Method	p	n	N	RE_{train}	RE_{test}
Linear	1	5	1	4.35×10^{-1}	4.54×10^{-1}
Quadratic	2	5	5	9.40×10^{-2}	$1. \times 10^{-1}$
Additive-AM	5	5	43	8.09×10^{-2}	8.54×10^{-2}
Sparse	5	5	43	1.35×10^{-2}	1.45×10^{-2}
Low-Rank	5	5	43	1.35×10^{-3}	1.65×10^{-3}
CPN-S ($\varepsilon = 10^{-4}$)	5	5	43	7.31×10^{-5}	7.51×10^{-5}

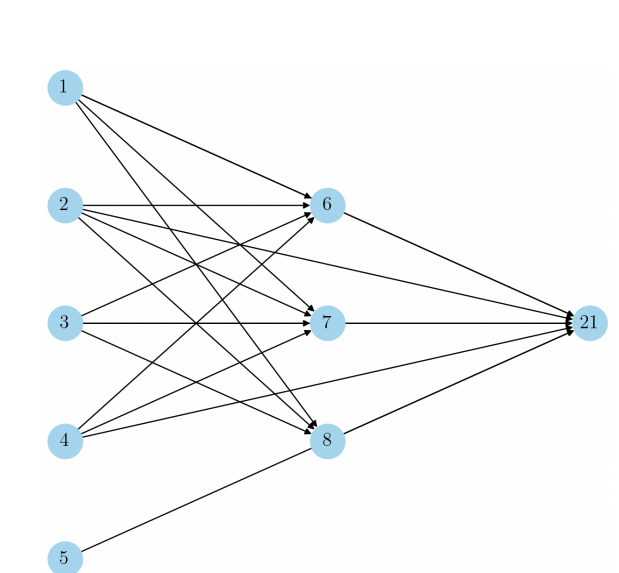
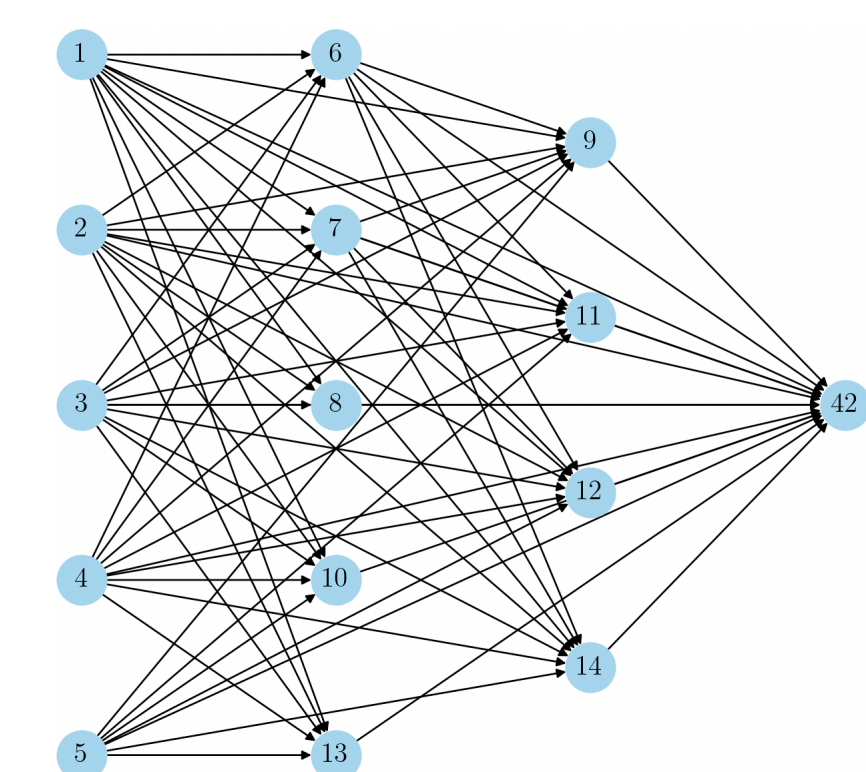
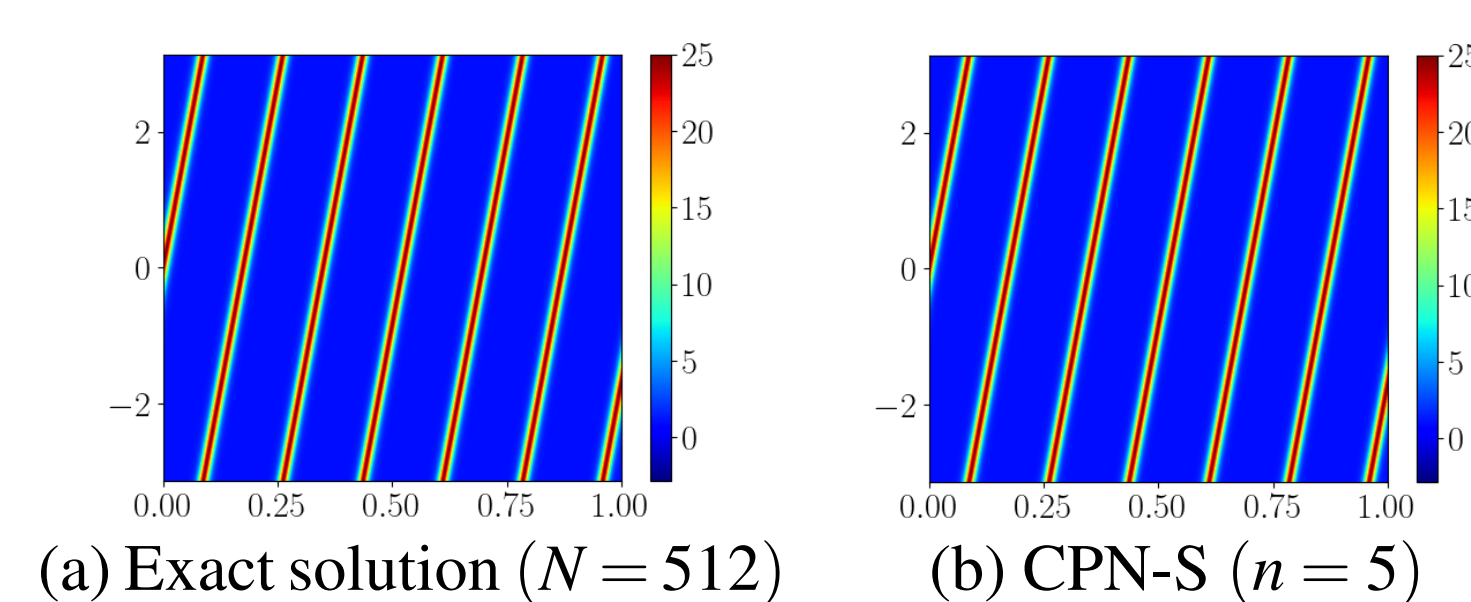
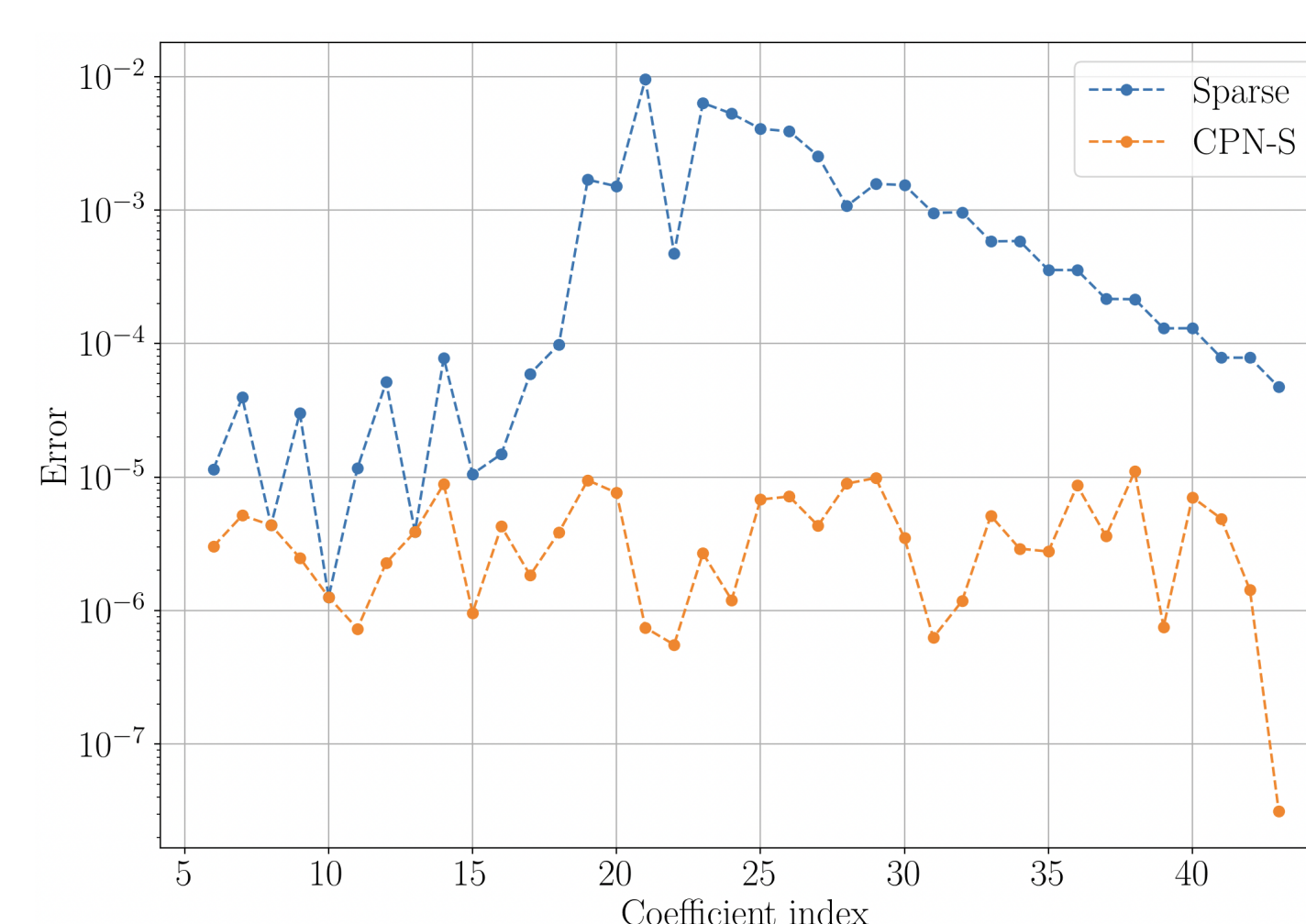


Figure 4: Compositional graphs for coeffs. 21 and 42.

Table 1: (KdV) Comparison of methods for the same manifold dimension $n = 5$.



KdV : Coefficient-wise errors



For $i \in \{6, \dots, 43\}$,

$$\text{Error}(i) := \frac{\sqrt{\sum_{k=1}^m |a_i^k - g_i(a^k)|^2}}{\sqrt{\sum_{k=1}^m \|u_k\|_X^2}}$$

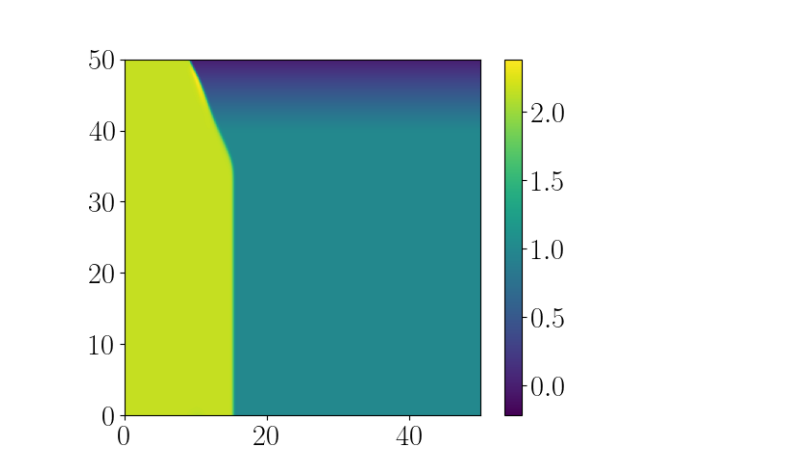
Burgers equation

$$\frac{\partial u_\lambda}{\partial t} + \frac{1}{2} \left(\frac{\partial u_\lambda^2}{\partial x} + \frac{\partial u_\lambda v_\lambda}{\partial y} \right) = 0, \quad \frac{\partial v_\lambda}{\partial t} + \frac{1}{2} \left(\frac{\partial u_\lambda v_\lambda}{\partial x} + \frac{\partial v_\lambda^2}{\partial y} \right) = 0.$$

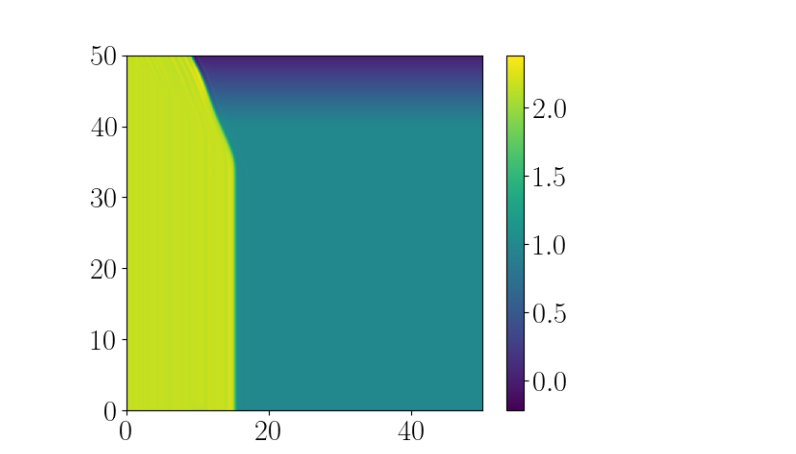
BCs : $u_\lambda(0, y, t) = \lambda, \quad t \in (0, 10).$
 $M = \{(u_\lambda, v_\lambda) : \lambda \in [1.5, 2.5]\}.$

Method	p	n	N	RE_{train}	RE_{test}
Linear	1	9	1	5.60×10^{-2}	5.01×10^{-2}
Quadratic	2	9	54	2.46×10^{-2}	2.28×10^{-2}
Additive-AM	8	9	89	/	/
Low-Rank	8	9	89	2.46×10^{-2}	2.33×10^{-2}
Sparse	8	9	89	1.62×10^{-2}	1.62×10^{-2}
CPN-S ($\varepsilon = 5.10^{-3}$)	8	9	89	4.74×10^{-3}	4.93×10^{-3}

Table 2: (2D Burgers) Comparison of methods for the same manifold dimension $n = 9$.



Exact solution ($\lambda = 1.157, t = 9.7$)



CPN-S ($n = 9, \lambda = 1.157, t = 9.7$)

Compositional Polynomial Networks

$$D(a) = \sum_{i \in I} a_i \phi_i + \sum_{i \in I^c} g_i(a) \phi_i, \quad a = E(u) = ((u | \phi_i))_{i \in I}, \quad I \cup I^c = \{1, \dots, N\}$$

$$\text{card}(I) = n$$

For $i \in I^c, g_i(a) = f_i((g_j(a))_{j \in S_i}), \quad S_i = \{1, \dots, n_i\}, n_i < i.$
 For $i \in I, g_i(a) = a_i$

N, I, n_i are selected by the adaptive algorithm.

Control of error

$$e_2(D \circ E)^2 := \sum_{k=1}^m \|u_k - D(E(u_k))\|_X^2.$$

Target precision : $0 < \varepsilon < 1$. It holds

$$e_2(D \circ E) \leq \varepsilon e_2(0)$$

when

$$e_2(D \circ E)^2 = \underbrace{\sum_{i=n+1}^N \|a_i - g_i(a)\|_2^2}_{\leq \left((1-\beta) \varepsilon e_2(0) \right)^2, 0 < \beta < 1} + \underbrace{\|id - P_{X_N}\|_2^2}_{\leq \left(\beta \varepsilon e_2(0) \right)^2}$$

Adaptive algorithm

- Prescribed precision ε , Lip. const L ,
- $\|id - P_{X_N}\|_2 \leq \beta \varepsilon e_2(0)$ is achieved through PCA,
- set $\varepsilon_i^2 = \omega_i(1 - \beta)^2 \varepsilon^2 e_2(0)^2, \tilde{\gamma}_i^2 = \omega_i(L^2 - 1),$
- Initialize $I = \{1\}, I^c = \{2, \dots, N\}, \forall i \in I^c, n_i = 0,$
- $\forall i \in I^c$, while $\|a_i - g_i(a)\|_2 > \varepsilon_i$ and $\gamma_i > \tilde{\gamma}_i$,
 Increase n_i until $n_i = i - 1$, then $I = I \cup \{i\}.$

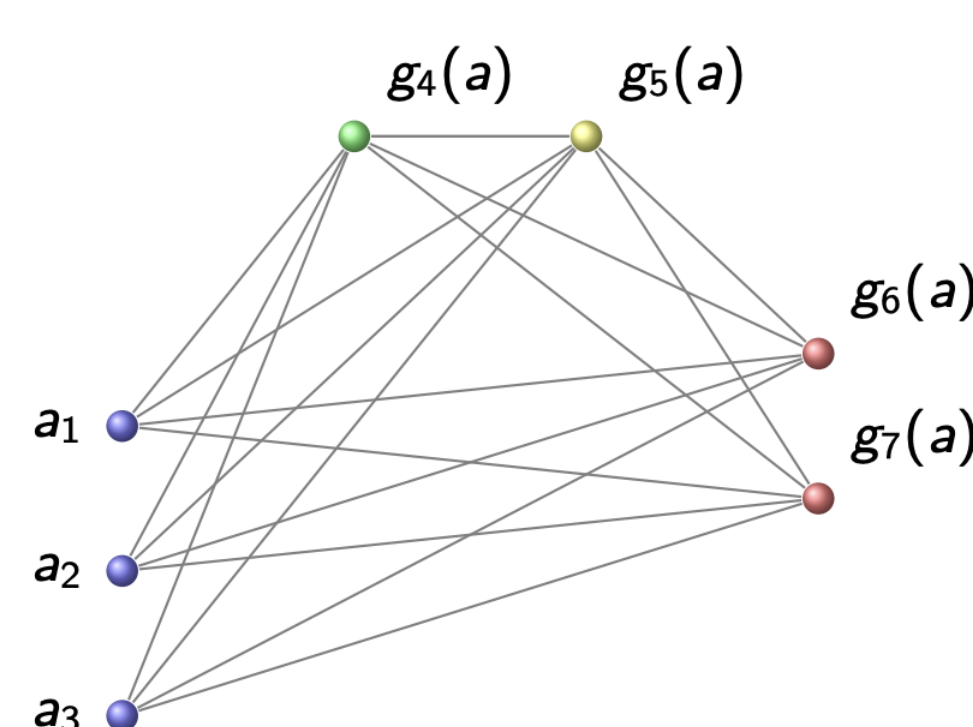


Figure 1: A CPN with $N = 7, n = 3(n_4 = 0, n_5 = 1, n_6 = 2, n_7 = 2).$

References

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Takeaways and next steps

- Adaptive strategy to construct an approximation manifold M_n with controlled error and stability,
- The dimensions n and N are found by the algorithm, depending on the prescribed precision ε and Lip. const. L ,
- The focus is now on using the proposed method for solving PDEs (efficient offline-online approximations).
- We are also working on operator learning methods using the proposed architecture.